

WASP-11b

R-2045

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_121117 · created 2026-08-08T23:34:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456244.95568 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.078 +/- 0.051
Transit depth (Rp/Rs)^2	0.6 +/- 0.79 [%]
Orbital Inclination (inc)	88.34 +/- 1.66
Airmass coefficient 1 (a1)	364.85 +/- 95.0
Airmass coefficient 2 (a2)	-3.0 +/- 1.28
Scatter in the residuals of the lightcurve fit is	19.97 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	3.0
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.078 $\pm$ 0.051 vs 0.131 (deviation 1.04 σ)
Duration fit vs published	0.0 d vs 0.1075 d (deviation inf σ)
Mid-time O-C	-5361.7 min
Depth SNR	1.5
Residual scatter	19.97 %

Light curve

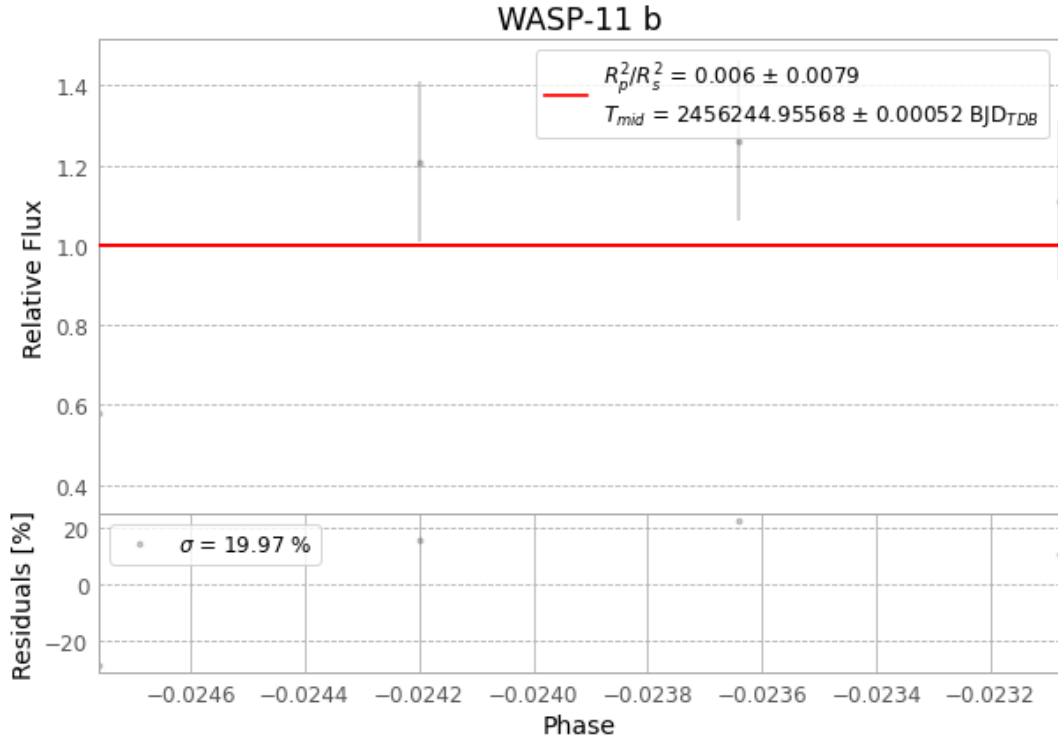


Figure 1 — FinalLightCurve_WASP-11 b_2012-11-16.png (SHA-256 4804fbc9d05a8dc0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [639, 161], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6e05e9d6e9ea4c3f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

93 FITS frames (61 MB), HATP-10121117015412.FITS → HATP-10121117063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-121117.log (bcf7dd2a4fd...), FinalLightCurve_WASP-11 b_2012-11-16.png (4804fbc9d05a...), FinalLightCurve_WASP-11 b_2012-11-16.csv (1bc8e725f1f8...)

Compute resources

Wall time 232.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:34Z.